From Planetary Chemistry to the Prevalence of Life

A Probabilistic Drake Framework

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Abstract

The biological factors of the Drake equation describe outcomes without specifying the processes that produce them. We develop a probabilistic framework in which those factors arise from transition rates, finite periods of opportunity, and variation among planets. For a single transition, the probability of completion is $1-\exp(-H)$, where H is the cumulative hazard. Under a specified model of candidate generation, H is the product of the number of opportunities and the probability that an opportunity establishes the required function. Sequential transitions are described by conditional waiting times rather than products of marginal probabilities; a common replication clock recovers the hard-steps approximation in its rare-event limit. Population averaging connects these transition models to expected counts, while an explicit observation model separates the presence of life from its detection. An illustrative RNA-replicase calibration places substantial prior weight on widespread abiogenesis and is considerably narrower than a diffuse prior on the transition rate. The adopted scenario gives a 0.77 prior probability that at least 10% of candidate worlds become inhabited. This is conditional on the chemical route, its establishment requirements, and the assumed planetary environments, rather than an empirical estimate of the probability that Earth is alone. The framework identifies the measurements needed to improve that estimate: functional frequencies under realistic chemistry, the production and turnover of candidate systems, the probability of sustained replication, and calibrated biosignature surveys. Its contribution is a mechanistic connection between chemistry, evolutionary opportunity, and observable populations, not a choice of logarithmic units.

1 Introduction

The Drake equation's enduring contribution is its decomposition of a difficult question. It separates the abundance of suitable astronomical environments from the biological transitions and observable lifetimes needed to produce detectable civilisations (Drake, 1965). Exoplanet surveys increasingly constrain the astronomical component, although a planet's radius and incident flux do not by themselves establish surface habitability. The biological factors remain much less constrained. Their uncertainty can dominate predictions even when the astronomical inputs are reasonably well characterised (Bryson et al., 2021; Sandberg et al., 2018).

A useful next step is therefore not to replace fractions with different units, but to explain where the fractions come from. The probability that a transition occurs depends on both the opportunities available and the chance that any one opportunity succeeds. A chemically difficult event can be common on planets that generate sufficiently many relevant candidates. Conversely, even an accessible function may never arise where its feedstocks, environmental conditions, or opportunities for establishment are absent. Subsequent evolutionary innovations introduce an additional constraint: opportunities for a later transition generally begin only after its predecessors have occurred.

We retain the idea that biological factors can be connected to searches through configurations, but use it as a modelling hypothesis rather than a universal description of evolution. A search model is useful when candidates, a sampling process, and a success criterion can be specified. It is less useful when those quantities conceal adaptive dynamics, long-lived correlations, or environmental changes that need to be modelled directly. The more general object is a *transition hazard*: the instantaneous rate of successful transition, conditional on its not yet having occurred.

This formulation links four parts of the problem. Laboratory experiments constrain functions and candidate-generation processes. Geochemistry and ecology constrain how opportunities accumulate. Waiting-time models describe ordered transitions. Population and observation models translate completed transitions into inhabited worlds and detectable signatures. The resulting framework preserves the Drake equation's decomposition while exposing the assumptions needed to turn it into an inference problem.

2 A probabilistic framework

2.1 From candidate generation to a transition probability

Let w describe a world's relevant physical conditions and let θ contain parameters shared across worlds. For transition k, let Ω_k be a space of candidate configurations, $F_k \subseteq \Omega_k$ the set satisfying a specified functional and establishment criterion, and q_k the distribution generating candidates. The per-opportunity success probability is

$$p_k(w, \theta) = q_k(F_k \mid w, \theta). \tag{1}$$

The criterion must match the event of interest. Producing a molecule with catalytic activity is not automatically the same event as establishing a self-maintaining hereditary system.

For A_k independent, identically distributed opportunities, the probability of at least one success is exactly $1 - (1 - p_k)^{A_k}$. If opportunities instead arrive as a Poisson process with rate $a_k(t)$ and are independently successful with probability $p_k(t)$, the successful-event rate is $h_k(t) = a_k(t)p_k(t)$. More generally, a model may specify h_k directly. If the transition becomes available at time s, its cumulative hazard and completion probability are

$$H_k(s, t \mid w, \theta) = \int_s^t h_k(u \mid s, w, \theta)\,\mathrm{d}u, \qquad P_k(s, t \mid w, \theta) = 1 - e^{-H_k(s,t|w,\theta)}. \tag{2}$$

Here the hazard is conditional on no intervening completion. The displayed notation assumes that conditioning on the preceding transition time and the specified world state suffices; additional relevant history must otherwise be retained.

For stationary independent opportunities, $H_k \simeq A_k p_k$. This is a rare-per-trial approximation for a fixed number of Bernoulli trials and an exact expression when the number of opportunities is Poisson with mean A_k. If the event permanently changes the system, H_k remains an integrated hazard; interpreting it as the number of realised transitions after the first event would be inappropriate.

Correlation cannot generally be repaired by counting each distinct sequence once. Repeated independent draws are valid opportunities, whereas correlated draws can have a different first-success distribution. A proposed effective trial count must be justified by the stochastic process or calibrated against its waiting-time distribution. Likewise, an uncertain random hazard must be averaged as $\mathbb{E}[1 - e^{-H}]$, not replaced by $1 - e^{-\mathbb{E}[H]}$.

Functional information can still serve as an auxiliary statistic. Under a specified sampling distribution, $-\log p_k$ measures the rarity of the functional set; under uniform sampling it reduces to the

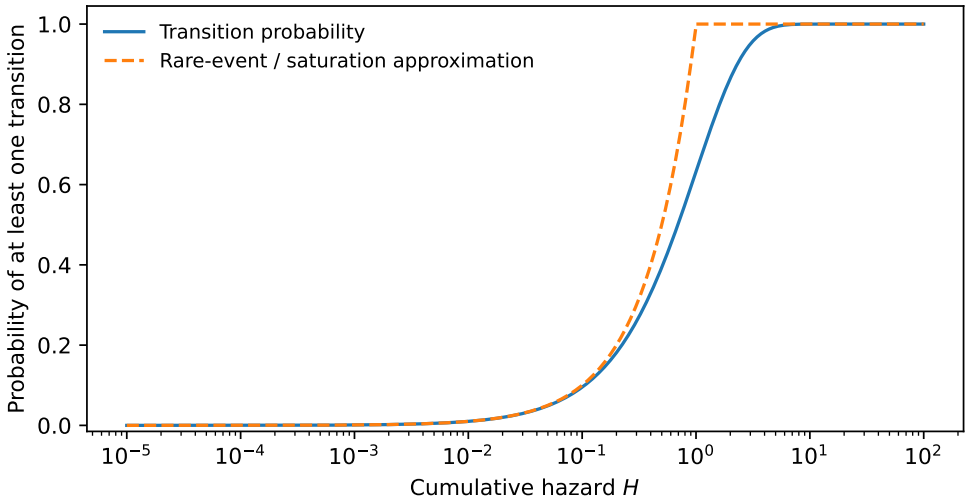


Figure 1: The transition probability as a function of cumulative hazard. The approximation $\min(1, H)$ captures the rare-event and saturation limits but is not used for the numerical population averages reported here. The horizontal axis describes an integrated transition rate, not an information budget.

familiar sequence-space definition (Hazen et al., 2007). The probability and its generating distribution are primary. Changing the logarithm's base does not change any prediction, and a rarity estimate under a laboratory distribution is not automatically valid under prebiotic synthesis.

2.2 Rare events and saturation

Equation (2) has two useful limits:

$$1 - e^{-H} \simeq \begin{cases} H, & H \ll 1, \\ 1, & H \gg 1. \end{cases} \tag{3}$$

A transition is unlikely when little cumulative opportunity has accumulated and nearly certain when the integrated success rate is large. Completion probabilities of 0.01 and 0.99 correspond to $H = 0.01005$ and $H = 4.605$, respectively: a factor of approximately 458 in cumulative hazard, or 2.66 orders of magnitude. This is a relatively narrow crossover when plausible hazards span many orders of magnitude, but it is not a physical discontinuity (Figure 1).

The useful logarithmic approximation is $-\log(1 - e^{-H}) \simeq [-\log H]_+$, with a uniform error bound given in Appendix A.1. It is a numerical convenience, not an additional law. In particular, the crossover does not imply that intermediate population prevalences require fine-tuning. Mixtures of very favourable and very unfavourable worlds can produce any intermediate fraction, and the conclusion also depends on the prior distribution of hazards.

2.3 Sequential transitions

Let T_k be the completion time of the kth required transition, with $T_0 = 0$, and let $g_k(t \mid w, \theta)$ be its density. For a serial model in which the previous completion time and the specified state contain the relevant history,

$$g_k(t \mid w, \theta) = \int_0^t g_{k-1}(s \mid w, \theta) h_k(t \mid s, w, \theta) e^{-H_k(s,t|w,\theta)} \, \mathrm{d}s, \qquad g_0(s) = \delta(s). \tag{4}$$

The probability of reaching state X by age T is the integral of the final-stage density up to T. Rates can be set to zero outside the relevant environmental windows. Competing pathways, reversals, or extinctions require a larger state model rather than an extra scalar correction.

A product of full-window probabilities generally gives each transition more time than it actually has. Even two transitions with the same constant rate h illustrate the distinction: their sequential completion probability is $1 - e^{-hT}(1 + hT)$, not $(1 - e^{-hT})^2$. Dividing the latter by two repairs only its leading rare-event term; it fails when both transitions become easy. The factorial hard-steps factor is therefore derived only under its own assumptions in Section 3.5, not appended to the general equation.

2.4 From worlds to expected populations

Let Z_X denote the random number of worlds reaching state X, and let $\mu_X = \mathbb{E}[Z_X]$ denote its expectation. Conditional on θ, a population of n_{hab} candidate worlds with characteristics drawn from $\pi(w \mid \theta)$ has

$$\mu_X^{\text{ever}}(T \mid \theta) = n_{\text{hab}} \int P_X(T \mid w, \theta)\pi(w \mid \theta)\, \mathrm{d}w. \tag{5}$$

The reference time and eligible population must be specified. A present-day stellar census is not itself a count of all habitable opportunities over Galactic history.

For a time-dependent formation history, let $b(t_b, w \mid \theta)\, \mathrm{d}t_b\, \mathrm{d}w$ be the expected number of candidate worlds born in the corresponding interval. The expected number currently in state X is

$$\mu_X^{\text{now}}(t_0 \mid \theta) = \int_0^{t_0} \int b(t_b, w \mid \theta)\mathbb{P}\{X \text{ is occupied at age } t_0 - t_b \mid w, \theta\}\, \mathrm{d}w\, \mathrm{d}t_b. \tag{6}$$

The occupancy probability includes the time to reach X and the duration or survival of that state. Under stationary formation over a sufficiently long history, this reduces to the Drake-like expression

$$\mu_X^{\text{now}} = R_{\text{hab}}\, \mathbb{E}_w\left[P_X^{\text{reach}}(w)\, \ell_X(w)\right], \tag{7}$$

where $\ell_X(w)$ is the mean total time spent in X conditional on reaching it. Replacing this expectation by a product of separate means additionally requires an independence assumption. Detectability belongs in an observation model unless X has explicitly been defined as a detectable state.

2.5 Planetary variation and uncertainty

Physical variation among planets and uncertainty about shared parameters are different levels of the model. At fixed θ, the inhabited fraction for a single origin transition is

$$f_{\text{life}}(\theta) = \mathbb{E}_w[1 - e^{-H_0(w,\theta)}]. \tag{8}$$

A prior or posterior over θ then induces uncertainty about this fraction. Averaging over w is not a substitute for learning θ, and uncertainty shared by all planets does not disappear in a large Galaxy.

For illustration, suppose $\log H_0 \sim \mathcal{N}(m, s^2)$ across worlds. In the deep rare-event limit at fixed s, $f_{\text{life}} \simeq \mathbb{E}[H_0] = e^{m+s^2/2}$. Relative to a homogeneous population with hazard equal to the *median* e^m, dispersion increases the mean hazard and can substantially increase the inhabited fraction. However, since $1 - e^{-H}$ is concave,

$$\mathbb{E}[1 - e^{-H}] \le 1 - e^{-\mathbb{E}[H]}. \tag{9}$$

Table 1: Astronomical assumptions used solely for the reference-population calculation. $\mathcal{U}$ denotes a uniform distribution; LogUnif is uniform in the logarithm. The normal distributions for $\log \eta$ use natural logarithms and variances as their second parameter.

Quantity	Adopted scenario
Galactic stars	$\mathrm{LogUnif}(10^{11}, 3 \times 10^{11})$
Observable-universe stars	$\mathrm{LogUnif}(10^{22}, 5 \times 10^{22})$
G, K, M shares	0.06, 0.12, 0.79; other classes omitted
GK occurrence	$\log \eta_{\mathrm{GK}} \sim \mathcal{N}(\log 0.30,\, 0.6^2)$
M occurrence	$\log \eta_{\mathrm{M}} \sim \mathcal{N}(\log 0.15,\, 0.8^2)$
Atmosphere-retention fractions	$f_{\mathrm{atm,GK}} \sim \mathcal{U}(0.3, 1)$; $f_{\mathrm{atm,M}} \sim \mathrm{LogUnif}(10^{-3}, 0.3)$

The population-scale assumptions draw on Bryson et al. (2021); Reylé et al. (2021); Licquia and Newman (2015); Frank and Sullivan (2016). The atmospheric distributions are additional scenario assumptions, not results inferred from these references. Distributions are independent here except where a parameter is explicitly shared. A formation-history calculation is needed for a count over all past opportunities.

At fixed arithmetic mean hazard, heterogeneity cannot increase the average success probability above the homogeneous case. There is no universal diversity bonus independent of the comparison being made.

Successful worlds are sampled according to $\pi(w \mid \mathrm{life}, \theta) \propto P_{\mathrm{life}}(w, \theta)\pi(w \mid \theta)$. They are consequently biased toward favourable conditions. An unusually productive prebiotic Earth is one possible implication if opportunity varies strongly between planets and dominates their success probabilities; it is not required when rarity instead reflects a shared chemical barrier or other selection effects. Appendix A.2 gives the lognormal calculation explicitly.

3 Constraining the transition model

3.1 The astronomical population

We write the abundance of candidate worlds schematically as $n_\star \sum_s x_s \eta_s f_{\mathrm{atm},s}$, where s indexes stellar classes, x_s is their share, η_s is the mean number of rocky habitable-zone planets per star, and $f_{\mathrm{atm},s}$ describes atmosphere retention. Occurrence rates constrain η_s, but incident flux and atmospheric retention are not complete habitability criteria (Bryson et al., 2021). Planet age, composition, water availability, and stellar history can affect both eligibility and transition hazards.

For a numerical reference population, we retain the scenario assumptions in Table 1. Independent propagation gives a Galactic candidate count with median 7.1×10^9 and central 68% interval $(3.4\text{–}14.7) \times 10^9$. Applying the same candidate fraction to an assumed $(1\text{–}5) \times 10^{22}$ stars gives a reference count of order 10^{21} for the observable universe. These are conditional population scales, not new empirical determinations of genuinely habitable worlds. In particular, the adopted atmospheric fractions are scenario priors, not measurements of all temperate rocky planets.

A population of 7.1×10^9 worlds has an expected one successful world when its average completion probability is approximately 1.4×10^{-10}. The corresponding threshold for the reference observable-universe population is of order 10^{-21}. This is the probabilistic content of the comparison between astronomical abundance and biological rarity (Frank and Sullivan, 2016); logarithmic units add no physical constraint. An expected count near one is a crossover in the count distribution, not a statement that exactly one world exists.

3.2 Abiogenesis: from catalytic activity to establishment

Laboratory selection provides a concrete way to study p_k under controlled conditions. A particularly relevant example is the QT45 RNA polymerase ribozyme. The published study demonstrated synthesis of its complementary strand and synthesis of the original strand in separate reactions; this establishes an important catalytic capability, not an autonomous, sustained replication cycle under uncontrolled planetary conditions (Gianni et al., 2026; MRC Laboratory of Molecular Biology, 2026). The earlier preprint estimated a sequence-space frequency of approximately 4.4×10^{-18} for its motif under a specified calculation (Gianni et al., 2024). Such estimates motivate route-specific priors, but do not directly measure the probability that a planet originates life.

For this route, at least three links must be made explicit. The first is the frequency of suitable molecular configurations under the relevant chemistry, rather than under an ideal random library. The second is whether compatible components meet with suitable substrates and cycling conditions. The third is whether their interaction establishes sustained heredity rather than a transient copying event. Performance, fidelity, degradation, product inhibition, compartment dynamics, and substrate supply can enter that last probability. They need not be independent.

A minimal compartment model makes the distinction quantitative. Suppose a compartment-generation event contains a Poisson number of functional molecules with mean $\lambda_c = n_c p_1$, where n_c is its expected candidate occupancy and p_1 the functional probability per candidate. The probability of containing at least ν functional molecules is

$$b_\nu(\lambda_c) = 1 - e^{-\lambda_c} \sum_{j=0}^{\nu-1} \frac{\lambda_c^j}{j!}. \tag{10}$$

Let K_c be the number of independent compartment-generation events and ρ the probability of establishment conditional on an event satisfying this occupancy requirement. For rare successful events,

$$H_0 \simeq K_c \, \rho \, b_\nu(n_c p_1) \simeq K_c \, \rho \, \frac{(n_c p_1)^\nu}{\nu!}. \tag{11}$$

The second approximation requires $n_c p_1 \ll 1$. If every event consumes a fresh group of n_c candidates, so that a total of A candidates corresponds to $K_c = A/n_c$, then for $\nu = 2$,

$$H_0 \simeq \frac{A \, n_c \, p_1^2 \, \rho}{2}. \tag{12}$$

This recovers the useful co-localisation scaling. It does not demonstrate that two randomly chosen active molecules are mutually compatible or sufficient for life. The parameter ρ represents precisely that missing connection, and must be defined to avoid double-counting requirements already included in p_1.

The mapping from A to K_c is a physical assumption, not a counting identity for arbitrary environments. Long-lived compartments, repeated encounters, unequal occupancies, and exchange between pores can alter it. Compartment volume and concentration should also be constrained jointly: allowing occupancy to increase while ignoring encounter times and dilution can create spurious gains.

3.3 Candidate production and an illustrative calibration

For a stationary oligomer-production scenario, a first estimate of the number of newly generated candidates is

$$A \simeq V c N_A \frac{T_{\mathrm{pre}}}{\tau}, \tag{13}$$

Table 2: Route-specific illustrative priors. Decimal logarithms are used only to express broad ranges compactly. Standard deviations are squared in the normal-distribution notation. The numerical implementation preserves unrounded endpoints; their provenance is documented in the accompanying revision notes.

Quantity	Adopted scenario
Functional motif frequency	$\log_{10} p_{\mathrm{fold}} \sim \mathcal{U}(-17.37, -13.55)$
Performance adjustment	$\log_{10} a_{\mathrm{perf}} \sim \mathcal{U}(-9.03, 0)$
Chemical sampling adjustment	$\log_{10} b_{\mathrm{chem}} \sim \mathcal{U}(-1.51, 3.01)$
Compartment occupancy	$\log_{10} n_c \sim \mathcal{U}(3.01, 9.03)$
Median candidate exposure across worlds	$\log_{10} \widetilde{A} \sim \mathcal{N}(36.12, 4.52^2)$
World-to-world exposure dispersion	$\mathrm{sd}(\log_{10} A \mid \theta) = 1.81$
Required occupancy; establishment	$\nu = 2$; baseline $\rho = 1$

The baseline assumes the listed quantities are independent at the parameter level and attributes interplanetary scatter to A alone. These choices simplify the illustration but are not implied by the underlying chemistry. $\rho = 1$ is an optimistic limiting case, not an experimental result.

where V is the active volume, c the molar concentration of candidates of the required kind, N_A Avogadro's constant, T_{pre} the available interval, and τ their replacement time. This is an exposure estimate. Its interpretation requires compatible definitions of production, turnover, and candidate identity. A general version integrates $V(t)c(t)N_A/\tau(t)$ over time, with zero contribution outside the relevant chemical conditions (Scharf and Cronin, 2016, for planetary-scale opportunity modelling).

To examine the consequences of a laboratory-motivated route, let $p_1 = p_{\mathrm{fold}} a_{\mathrm{perf}} b_{\mathrm{chem}}$. Here $a_{\mathrm{perf}} \leq 1$ is a conditional reduction for the specified performance requirement and b_{chem} is a multiplicative adjustment for the sampling chemistry. Table 2 specifies an illustrative scenario in probabilities and opportunity counts. The entries are assumptions informed to different degrees by experiments; they are not a jointly measured posterior. In particular, a performance–rarity relation observed for other RNA functions is not established for autonomous replication (Carothers et al., 2004).

For the median world in a parameter draw, these choices give $\log_{10} \widetilde{H}_0 \simeq 3.4 \pm 7.9$ before any additional establishment penalty. The comparison prior is specified directly as $\log_{10} H_0 \sim \mathcal{N}(0, 50^2)$, with the same hazard on every world conditional on the draw, following the broad rate uncertainty used by Sandberg et al. (2018). A diffuse prior should be placed on a nonnegative hazard, not implemented through a putative per-trial probability that can exceed one. The route-specific log-hazard dispersion is approximately 6.3 times smaller (Figure 2). That narrowing is conditional on choosing the route and its parameter ranges; it is not equivalent to a sixfold reduction achieved by new data.

3.4 Heredity and later evolutionary transitions

The ability to maintain heredity constrains which catalytic systems can establish. In the simple single-peak replication model, a master sequence with error-free copying probability Q and relative replication advantage σ requires $Q\sigma > 1$. With independent per-site error probability ϵ across L relevant sites, $Q = (1 - \epsilon)^L$ (Eigen, 1971). This is a useful consistency check on a proposed replicator.

It is not a universal upper bound on the negative logarithm of functional sequence frequency. Functional sets can contain neutral networks and tolerate many sequence changes; copying fidelity, mutational robustness, and function rarity are distinct properties. Establishment should be assessed with a model of the actual genotype–phenotype and replication dynamics rather than by converting every constrained site into a fixed information cost. Likewise, failure of one copying mechanism does not show that all excess functionality must arise in a single random assembly event.

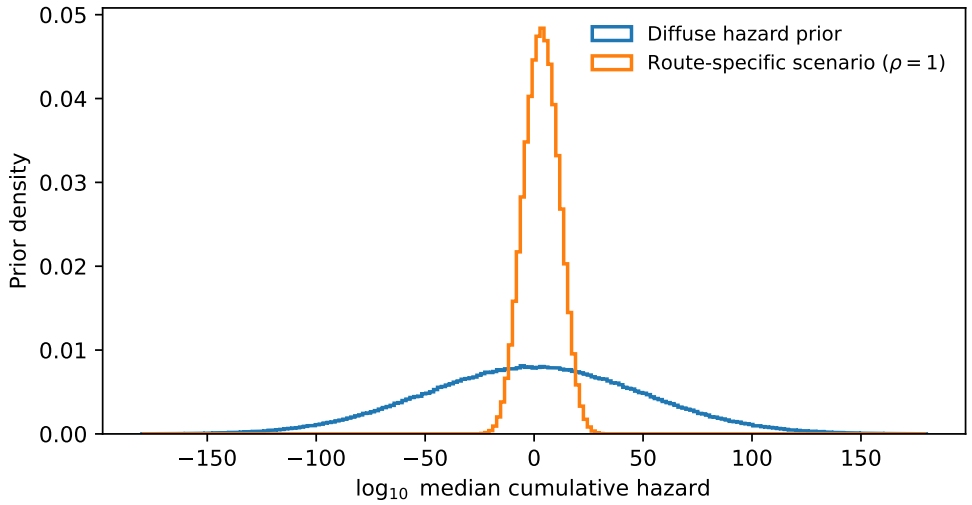


Figure 2: The route-specific prior on the median cumulative hazard compared with a diffuse hazard prior. Both distributions concern uncertainty in a model parameter. The separate world-to-world dispersion is integrated when computing the inhabited fraction. Narrower prior support is not, by itself, stronger empirical evidence.

After heredity exists, a transition hazard can depend on mutation, selection, population structure, recombination, ecological interactions, and the persistence of intermediate genotypes. Counting all cells that have ever existed provides neither the number of eligible trials for every innovation nor a universal bound on the number of coordinated mutations evolution can achieve. Fitness-valley crossing, for example, depends on the dynamics of intermediates as well as on individual mutation probabilities (Weissman et al., 2009). The opportunity-rate description remains useful when those dynamics can be reduced to a justified effective hazard.

3.5 The replication-clock hard-steps limit

Suppose n ordered transitions have hazards $h_k(t) = \alpha_k r(t)$ after their predecessors have occurred, where the same nonnegative opportunity clock $r(t)$ applies to every step. Define cumulative opportunities $U(T) = \int_0^T r(t)\,\mathrm{d}t$. If $\alpha_k U(T) \ll 1$ for every required transition, then

$$P_n(T) \simeq \frac{U(T)^n}{n!} \prod_{k=1}^{n} \alpha_k. \tag{14}$$

Conditional on completion, the coordinates $U(T_1), \ldots, U(T_n)$ have, to leading order, the distribution of the ordered values of n independent uniforms on $[0, U(T)]$. Appendix A.3 gives the derivation. Constant r recovers the familiar T^n hard-steps scaling (Carter, 1983).

This result preserves two important conclusions. Available biological opportunity, not calendar time alone, can control transition probabilities. Also, conditional timing in this rare-event model does not identify the individual rate scales α_k: their common multiplicative contribution cancels when conditioning on completion. Earth-like chronologies can therefore be compatible with very different unconditional probabilities of reaching observers. Inferring even the number of steps requires assumptions about the opportunity clock and the transitions included.

Environmental gating and intrinsic rarity can coexist, but one does not generally reduce to the other.

Table 3: Recomputed prior-predictive summaries for the reference population. $\mu = n_{\text{hab}} f_{\text{life}}$ is an expected count over the scenario's specified opportunities. The final column is prior mass below a mean-count threshold, *not* the probability that Earth is alone. Results use 10^6 parameter draws.

Scenario	$\mathbb{P}(\widetilde{H}_0 > 1)$	$\mathbb{P}(f_{\text{life}} \geq 0.1)$	$\mathbb{P}(\mu < 1)$
Route-specific, $\rho = 1$	0.661	0.772	0.013
Same priors, $\rho = 10^{-3}$	0.520	0.646	0.037
Same priors, $\rho = 10^{-6}$	0.377	0.505	0.084
Same priors, $\rho = 10^{-9}$	0.247	0.362	0.161
Same priors, $\rho = 10^{-12}$	0.144	0.234	0.271
Diffuse hazard comparator	0.500	0.508	0.422

A more flexible model is $h_k(t) = r_k(t)\alpha_k(t)g_k(t)$, where g_k describes the environmental permissibility of step k. A gate may prevent a transition even while unrelated organisms are abundant. The common-clock theorem applies only when these effects can actually be represented by proportional rates. Otherwise Equation (4) or a corresponding state model is needed. This gives a common framework for comparing hard-step and environment-gated accounts without claiming that they are observationally equivalent (Mills et al., 2025).

4 Consequences and observational tests

4.1 What the illustrative abiogenesis model implies

An independent Monte Carlo propagation of Tables 1 and 2, with numerical averaging of Equation (8), gives $\mathbb{P}(\widetilde{H}_0 > 1) = 0.661$ and $\mathbb{P}(f_{\text{life}} \geq 0.1) = 0.772$ for the baseline scenario. The median expected count in the reference population is approximately 4.3×10^9. The diffuse hazard prior gives $\mathbb{P}(f_{\text{life}} \geq 0.1) = 0.508$. Thus the route-specific assumptions favour widespread abiogenesis relative to the diffuse comparator, without making that conclusion certain.

Table 3 exposes a missing sensitivity: establishment after co-localisation. Decreasing ρ shifts $\log_{10} H_0$ by $\log_{10} \rho$. The example values are stress tests, not estimated probabilities or a confidence range. More generally, changing pathways, correlating chemical inputs, adding missing prerequisites, or truncating the favourable environmental tail can alter the inference beyond one-at-a-time variations of existing parameters.

The supported conclusion is conditional but substantive: a specified small-replicase route does not require an extremely low planetary success probability under these opportunity and establishment assumptions. The stronger assertion that laboratory measurements already establish a small probability of Galactic solitude does not follow. That claim would require both a defensible likelihood for the chemical parameters and an explicit treatment of how our own existence enters the data.

4.2 Expected counts, solitude, and observer selection

For conditionally independent worlds with completion probabilities $f_j(\theta)$,

$$\mathbb{P}(Z_X = 0 \mid \theta) = \prod_j [1 - f_j(\theta)], \qquad \mu_X(\theta) = \sum_j f_j(\theta). \tag{15}$$

When individual probabilities are small, the zero-count probability is approximately $e^{-\mu_X}$. At $\mu_X = 1$, this is about 0.37, not certainty of either solitude or company. Shared uncertainty requires integrating

Equation (15) over the parameter distribution; exponentiating a posterior mean count is generally incorrect.

If $\mathcal{D}$ includes the relevant terrestrial and observational evidence, the probability of no other inhabited worlds is

$$\mathbb{P}(Z_{\text{other}} = 0 \mid \mathcal{D}) = \int \prod_{j \neq \oplus} [1 - f_j(\theta)]\, p(\theta \mid \mathcal{D})\, \mathrm{d}\theta. \tag{16}$$

A probability that μ is below one, or below two, answers a different question. Nor can a known inhabited Earth simply be subtracted from an unconditioned expected population without updating the model.

Conditioning on at least one inhabited world somewhere in the observable universe is one possible selection event, not a unique prescription for conditioning on our observations. The relevant likelihood depends on what is sampled: planets, inhabited histories, observer-producing histories, or observers. Earth's early evidence for life also enters through uncertain habitability and detection times, together with the time needed for observers to emerge (Spiegel and Turner, 2012; Kipping, 2020). The illustrative prior propagation above is deliberately not presented as this posterior calculation. Appendix A.4 gives a simple conditional-timing likelihood to make the distinction explicit.

4.3 Host stars and technological states

The replication-clock limit supplies a conditional prediction for host-star statistics. If comparable worlds around different stellar classes have the same per-opportunity transition coefficients, and differ mainly in usable opportunity U_s, the production of observer-bearing histories scales as U_s^n in the rare-event regime. Additional hard steps therefore strengthen the preference for stellar populations offering larger accumulated opportunity. A G-dwarf host can favour fewer such steps when the comparison population of longer-lived stars has comparable chemistry, productivity, and observer weighting.

A quantitative likelihood nevertheless requires the distribution of stellar formation times, age-limited habitable windows, productivity, and a definition of the observer sample. Counting worlds that ever produced observers is not the same as counting observers alive today. Host luminosity or spectral class can change both environmental hazards and detectability. Bayes factors for ranges of step counts further require priors within those ranges and over nuisance parameters. The host-star argument is consequently a useful proposed test, not a model-independent measurement of the number of hard steps (Haqq-Misra et al., 2018). An appropriate weighting expression is given in Appendix A.5.

For technology, the same distinction between occurrence and occupancy applies. Under stationary births and a representative constant lifetime ℓ, a reference population formed approximately uniformly over T_{pop} would have $\mu_{\text{tech}}^{\text{now}} \simeq n_{\text{hab}} \overline{P}_{\text{tech}} \ell / T_{\text{pop}}$. Taking $n_{\text{hab}} \simeq 7 \times 10^9$ and $T_{\text{pop}} = 10^{10}$ yr solely as an illustration, a mean contemporary count of one requires $\overline{P}_{\text{tech}} \simeq 1.4 \times 10^{-4}$ for $\ell = 10^4$ yr, or 1.4×10^{-6} for $\ell = 10^6$ yr. These are conditional rate–lifetime tradeoffs, not sharp boundaries between possible and impossible civilisations.

Non-detections constrain the abundance of sources only through the surveys' selection functions. A geometric fraction of a proposed SETI search space is not automatically the probability of detecting a civilisation; the distribution of sources within that space matters (Wright et al., 2018). A single detection would be transformative, but would not independently identify evolutionary probabilities, emission properties, and lifetimes.

4.4 Biosignatures as model comparison

The likelihood that a process produces a functional system and the likelihood that it produces an observation are related modelling problems, not generally the same probability. Let $\mathcal{L}$ and $\mathcal{A}$ denote biotic and abiotic hypotheses, and let d be the observed data. The evidential comparison is the Bayes factor

$$\mathrm{BF}_{\mathcal{L},\mathcal{A}}(d) = \frac{\int p(d \mid \psi_{\mathcal{L}}, \mathcal{L})\pi(\psi_{\mathcal{L}} \mid \mathcal{L})\,\mathrm{d}\psi_{\mathcal{L}}}{\int p(d \mid \psi_{\mathcal{A}}, \mathcal{A})\pi(\psi_{\mathcal{A}} \mid \mathcal{A})\,\mathrm{d}\psi_{\mathcal{A}}}. \tag{17}$$

Posterior odds equal prior odds times this factor. For continuous observations the quantities integrated are likelihood densities; the probability of one exact measured value is not the relevant event probability.

There is a useful special connection to the transition model. If a pre-specified signature event E is triggered by at least one abiotic production event with Poisson mean $H_{\mathcal{A}}$, then $\mathbb{P}(E \mid \mathcal{A}) = 1 - e^{-H_{\mathcal{A}}}$. For $H_{\mathcal{A}} \ll 1$, $\log \mathrm{BF}(E) \simeq -\log H_{\mathcal{A}} + \log \mathbb{P}(E \mid \mathcal{L})$. This expresses the sound underlying idea: a persuasive signature must be difficult for a realistic abiotic model to generate and plausible under a biotic alternative. The Poisson representation is an additional model of the observation, however, not a universal identity for spectra or molecular complexity.

Atmospheric composition, molecular ensembles, disequilibria, and temporal variability should therefore be evaluated through competing generative models. A gas is not intrinsically weak evidence merely because it is a single species, nor is a complicated pattern intrinsically strong evidence if abiotic alternatives or measurement systematics predict it. Statistical significance is also not a model-independent conversion to evidence for life: a p-value is not a Bayes factor, and calibrations require stated assumptions (Sellke et al., 2001). The same caution applies to molecular-complexity thresholds and to technosignatures. A thermodynamic dissipation bound, for example, becomes a detection prediction only after specifying a process rate, temperature, and observable emission model (Landauer, 1961).

4.5 What surveys and experiments can establish

For target i, let $f_i(\theta)$ be the probability of life in the specified sense, d_i the probability of a positive classification given life, and α_i the probability of a positive classification without life. The detection efficiency can include whether an inhabited planet currently produces a detectable signature. Then

$$q_i(\theta) = \alpha_i + (d_i - \alpha_i) f_i(\theta), \qquad Y_i \sim \mathrm{Bernoulli}\big(q_i(\theta)\big). \tag{18}$$

The joint likelihood is the product over targets conditional on shared parameters; uncertainty in calibration or population structure must then be integrated. Unknown efficiency can be confounded with prevalence, and target selection must be included when the surveyed planets do not represent the population of interest.

A few dozen well-characterised planets can strongly distinguish nearly ubiquitous life from extremely rare life under suitably concentrated priors. The crossover in Equation (3) helps explain why such priors arise in some chemical models, but does not by itself guarantee a decisive survey.

For a specific forecast, take independent targets with known $d = 0.6$ and $\alpha = 0.02$, and define $S = \{f_{\mathrm{life}} \geq 0.1\}$. Under the route-specific baseline, 25 targets reduce the expected uncertainty about S by 0.449 nats out of a prior entropy of 0.537 nats, and give a posterior probability above 0.95 or below 0.05 in 84% of prior-predictive outcomes. Five targets achieve that decision threshold in 62% of outcomes (Table 4). These forecasts support the value of modest surveys within the specified model. They do not include unknown shared calibration errors or establish a prior-independent sample-size guarantee.

Table 4: Recomputed survey forecasts under Equation (18) with known detection efficiency 0.6 and false-positive rate 0.02. The target proposition is $S = \{f_{\text{life}} \geq 0.1\}$. Entropy reduction uses natural logarithms; "resolves S" means a posterior probability at least 0.95 or at most 0.05. Values are conditional on each stated prior and this observation model.

Prior	Targets	Entropy reduction (nats)	Probability of resolving S
Route-specific baseline	5	0.328	0.615
Route-specific baseline	25	0.449	0.838
Diffuse hazard comparator	5	0.582	0.912
Diffuse hazard comparator	25	0.678	0.990

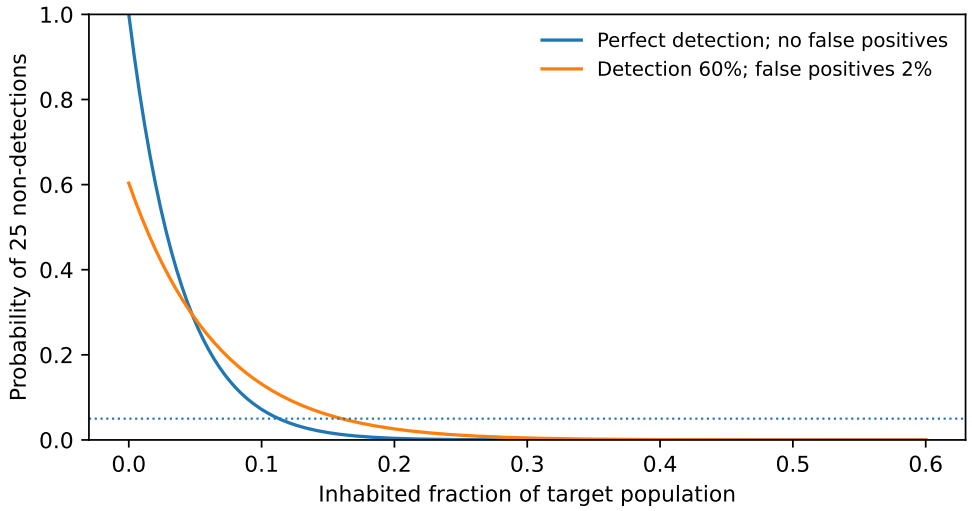


Figure 3: The likelihood of 25 non-detections under two explicitly calibrated observation models. The horizontal dotted line marks a probability of 0.05. A small survey's ability to establish a prevalence threshold depends on efficiency, false positives, the population model, and prior uncertainty, not just the shape of the single-world transition curve.

The dependence on the inferential question is important. For 25 independent targets with perfect detection and no false positives, zero detections have probability $(1 - f)^{25}$. At $f = 0.1$ this is 0.0718, and the exact one-sided 95% frequentist upper limit after zero detections is $1 - 0.05^{1/25} = 0.113$. Thus 25 non-detections do not quite exclude a prevalence of 10% even in this ideal case. With $d = 0.6$ and $\alpha = 0.02$, their probability at $f = 0.1$ rises to 0.131 (Figure 3). These are likelihood calculations, not posterior conclusions.

Laboratory and geochemical measurements address complementary factors in H_0. Functional screening under realistic synthesis conditions constrains candidate probabilities. Measurements of oligomer abundance and turnover constrain exposure. Encapsulated, sustained replication constrains establishment and the relation between occupancy and successful heredity. Multiple plausible chemical routes should be tested rather than treating one as exhaustive. Route hazards can be added when they are distinct competing event channels in the same eligible state; simply adding probabilities of overlapping functional sets double-counts configurations.

A genuinely independent second origin would provide a particularly important likelihood contribution, especially in an environment predicted to offer fewer opportunities. It would not impose a deterministic lower bound on every world's success probability or guarantee that life is common: rare

events can occur, routes can differ, and shared ancestry or transfer must be excluded (Spiegel and Turner, 2012).

Experiments can be compared by their expected reduction in uncertainty about a stated proposition, such as $S = \{f_{\text{life}} \geq 0.1\}$. For design a, one suitable quantity is $\mathrm{H}(S \mid \mathcal{D}) - \mathbb{E}_{y|a,\mathcal{D}}[\mathrm{H}(S \mid \mathcal{D}, y, a)]$, where H is entropy. This is a genuine information-theoretic quantity, but its numerical value depends on the prior, the experiment's likelihood, and the decision threshold. A claim that a particular programme will settle the prevalence of life requires those assumptions and a reproducible calculation, rather than the transition crossover alone.

5 Discussion

The framework is deliberately modular. A mechanistic model can refine a transition hazard without changing the population accounting; a new survey can improve the observation model without pretending to determine the chemical mechanism uniquely. This separation also clarifies identifiability. A measured transition fraction often constrains a cumulative hazard but cannot distinguish many low-probability opportunities from fewer high-probability ones. Laboratory data and environmental measurements are needed to separate those contributions.

Three limits deserve emphasis. First, a convenient search model is not evidence that all relevant chemistry is independent sampling. Persistent networks, spatial correlations, changing feedstocks, and prior selection can change both the generated candidates and their chances of establishment. Second, a chosen functional route is not the full event "life originates". Alternative routes can raise the total hazard, while missing prerequisites can lower it; both are structural uncertainties. Third, a planet is not an isolated Bernoulli trial if material transfer, common formation history, or shared environments create dependence. Independence is a conditional modelling assumption whose adequacy depends on the scale of the population.

The framework builds directly on existing work rather than claiming a new probability law. The single-event expression is the standard waiting-time relation used in abiogenesis models (Sandberg et al., 2018); planetary exposure modelling connects it to physical opportunities (Scharf and Cronin, 2016). The replication-clock limit extends the interpretation of hard-step timing without removing its assumptions (Carter, 1983; Mills et al., 2025). Functional rarity remains useful when its sampling measure and success criterion are specified (Hazen et al., 2007). The astronomical abundance threshold is the inverse of a candidate-population size (Frank and Sullivan, 2016). Their integration is valuable because it turns informal biological fractions into quantities linked to explicit mechanisms, data, and tests.

The illustrative RNA calculation sustains the motivating optimism: widespread abiogenesis is compatible with a concrete small-replicase route and substantial planetary opportunity. Its uncertainty is narrower than that of a deliberately diffuse rate model. The narrowing should not be oversold. It arises partly because the chemical route, parameter support, and missing establishment probability have been fixed. Progress means measuring those assumptions, comparing alternative mechanisms, and reporting how posterior predictions change, not merely choosing a tighter prior.

6 Conclusion

A biological factor in the Drake equation can be treated as the probability that a specified transition completes within a finite period of opportunity. The central quantitative object is the cumulative hazard, built where possible from candidate production, functional probability, and successful es-

tablishment. Ordered transitions require conditional waiting times; observable populations require formation histories, occupancy durations, and detection models.

This perspective preserves the useful conclusions of an opportunity-based account. Astronomical abundance can compensate for low per-world probabilities. Favourable planets can contribute disproportionately, although the effect of heterogeneity depends on the comparison population. Evolutionary timing may constrain aspects of an opportunity clock while leaving absolute transition rates weakly identified. Laboratory results can motivate substantially more informative chemical models, and modest, well-calibrated surveys can be highly discriminating.

The remaining uncertainty is attached to identifiable questions: which chemical systems can establish heredity, how often they are generated in realistic environments, how much relevant opportunity planets provide, and which observations distinguish life from abiotic alternatives. The purpose of the framework is not to declare those questions settled. It is to make their assumptions explicit and connect their answers to the prevalence of observable life.

Code and numerical scope. The accompanying `recompute.py` independently implements the illustrative priors, numerical population averages, reference counts, establishment stress tests, and three figures in this revision. The reported run uses 10^6 draws and seed 20261014; the quadrature is checked against adaptive integration. The code also computes Table 4 under its explicitly known survey calibration. It does not estimate an observer-conditioned solitude posterior, a host-star Bayes factor, or laboratory-experiment utilities; those require additional likelihoods and assumptions.

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A Technical details

A.1 A bound for the logarithmic approximation

For $H \geq 1$, $1 - e^{-H}$ increases from $1 - e^{-1}$ to one. For $0 < H < 1$, $(1 - e^{-H})/H$ decreases from one to $1 - e^{-1}$. Hence

$$(1 - e^{-1}) \min(1, H) \leq 1 - e^{-H} \leq \min(1, H), \tag{19}$$

which gives

$$0 \leq -\log(1 - e^{-H}) - [-\log H]_+ \leq -\log(1 - e^{-1}) \simeq 0.459. \tag{20}$$

The logarithm is natural here. Under a justified product model these errors add across factors, but the bound does not control the error from replacing a sequential process by such a product. The exact Bernoulli exponent is $-A \log(1 - p) = Ap\{1 + p/2 + O(p^2)\}$; thus the Poisson replacement has a relative exponent error of order p.

A.2 The lognormal population average

Let $X = \log H \sim \mathcal{N}(m, s^2)$, $s > 0$. Replacing the exact success function by $\min(1, e^X)$ gives

$$\mathbb{E}[\min(1, e^X)] = \Phi\left(\frac{m}{s}\right) + e^{m+s^2/2}\Phi\left(-\frac{m + s^2}{s}\right). \tag{21}$$

This follows by completing the square in the truncated normal moment. It is an upper approximation to the exact inhabited fraction, with a universal multiplicative lower bound obtained from Equation (19). Numerical calculations in the main text use the exact expectation instead. In the limit $m \to -\infty$ at fixed s, saturation becomes negligible in the relevant part of the distribution and the result tends to $e^{m+s^2/2}$.

In the same rare-event limit, weighting worlds by success multiplies their density by e^X. Consequently, X among successful worlds is approximately $\mathcal{N}(m + s^2, s^2)$. This is a conditional sampling bias, not proof of an anomalous Earth. Its applicability depends on the assumed source of interplanetary variation, the accuracy of the rare-event limit, and any additional observer selection. Predictions dominated by an extrapolated upper tail should be checked against physical truncations and finite-population sampling.

A.3 Ordered completion on a common clock

Set $u = U(t)$ and $U = U(T)$. In this coordinate, stage k has constant hazard α_k after stage $k-1$. The joint completion density on $0 < u_1 < \cdots < u_n < U$ is

$$\prod_{k=1}^{n} \alpha_k \exp\{-\alpha_k(u_k - u_{k-1})\}, \qquad u_0 = 0. \tag{22}$$

When $\sum_k \alpha_k U$ is small, replacing the exponential factors by one has a relative error of that order. The ordered simplex has volume $U^n/n!$, yielding Equation (14). Normalisation by the completion probability removes $\prod_k \alpha_k$ and gives the uniform order-statistics density $n!/U^n$.

For equal constant hazards h, the exact completion probability for n stages is the Erlang cumulative distribution,

$$P_n(T) = 1 - e^{-hT} \sum_{j=0}^{n-1} \frac{(hT)^j}{j!}. \tag{23}$$

It approaches $(hT)^n/n!$ only in the rare-event limit and approaches one when hT is large. For step-specific opportunity clocks, the leading rare-event result is instead an ordered integral of $\prod_k h_k(t_k)$. It does not generally factor into a product of full-window hazards divided by $n!$.

A.4 A simple observer-conditioned timing likelihood

Assume, solely for illustration, that abiogenesis has constant hazard h from time zero, that the observation establishes completion by t_e, and that observers can exist only if completion occurs by a known deadline $t_{\max} \geq t_e$. If that deadline is the entire selection model, the likelihood of the early-completion observation conditional on selection is

$$\mathbb{P}(T_{\text{life}} \leq t_e \mid T_{\text{life}} \leq t_{\max}, h) = \frac{1 - e^{-ht_e}}{1 - e^{-ht_{\max}}}. \tag{24}$$

As $h \to 0$ this tends to $t_e/t_{\max}$, rather than zero; for very large h it tends to one. The evidence is therefore limited under this selection model. Uncertain dates, variable hazards, and stochastic later evolution require integration over the appropriate additional variables (Spiegel and Turner, 2012; Kipping, 2020).

A.5 Host-star likelihoods

Let $b_s(t_b)$ describe eligible world births around stellar class s, $g_{n,s}(u \mid \psi)$ the density of first observer-producing completion at planetary age u, and $\omega_s(t_0, t_b + u, \psi)$ the expected contribution of that completed history to the specified observer sample. Then a general weight is

$$W_s(n, \psi) = \int_0^{t_0} b_s(t_b) \int_0^{t_0 - t_b} g_{n,s}(u \mid \psi) \omega_s(t_0, t_b + u, \psi) \, \mathrm{d}u \, \mathrm{d}t_b, \quad \mathbb{P}(s \mid n, \psi, \mathcal{O}) = \frac{W_s}{\sum_{s'} W_{s'}}. \tag{25}$$

Setting $\omega_s = 1$ counts successful world histories by the present, not all observers currently alive. Observation at a particular epoch generally needs a duration-dependent weight. Environmental gates and age limits enter $g_{n,s}$; spectral effects on productivity enter the opportunity clock. A comparison of composite hypotheses such as $n \leq 2$ and $3 \leq n \leq 6$ must average the host likelihood over specified within-hypothesis priors for n and ψ. These requirements prevent a conditional host preference from being mistaken for a universal Bayes factor.

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